# Detecting Adverse Drug Effects from Genomes of Patients with Cancer

*Introduction*

The scope of this project is to create a high-performing data science based solution that can predict cancer drug sensitivity based on genetic features. This insight will allow healthcare professionals to provide more information on how individual patients may respond to specific cancer drugs and overall improve treatment effectiveness while reducing adverse side effects provided from inaccurate personalized treatments.

In a previous study done by Larisa H. Cavallari and Julie A. Johnson investigating the *Use of a multi-gene pharmacogenetic panel reduces adverse drug effects*. Focusing on how using a multi-gene pharmacogenetic panel can assist in prescribing for healthcare professionals to reduce relevant adverse drug reactions in patients. This study was different because previous studies were limited to single drug-gene interactions. They looked at a larger scale of data to help benefit over 90% of individuals carrying an actionable pharmacogenetic variant, finding 50 genetic variants across 12 genes actionable in drug metabolism phenotypes. Similarly, the focus of the following project is the different drug relations between patients and adverse side effects. In the paper, they focused the metrics of their results closely on an odds ratio approach combined with p-value, whereas we took a pivot and chose LN_IC50 as our target variable which is the natural log of the drug concentration needed to reduce 50% of the tumor/cancer. This study gave us the framework. The purpose of our final project is to improve effectiveness in cancer drug prescription based on genetic features.

As a team we have uncovered an understanding of variability in cancer drug response and focused on our goal to match treatments to biological variables for different tumors. A scientific paper by Mathew Garnett et al. provided baseline information of the connection between genomic features of cancer cell lines and measured responses to anticancer compounds. Using this information to better evaluate our model. Our data sets provided great biological data but suffered from large amounts of missing values and inconsistencies. This made data cleaning and preprocessing vital to withdrawing important findings. We constructed a cleaned modeling dataset that was optimal for predicting drug response using LN_IC50 as our target variable. Through targeted data recovery, integration of supplementary biological features from Cell Lines

Encyclopedia and structured feature selection. Our efforts were not only to predict model performance but to provide biological interpretability. After cleaning, the primary model of the project focuses on ElasticNet regression and LightGBM using the most accurate representation of the true biological data. The original approach of the project was with LightGBM, which is a gradient boosting approach, is efficient and uses techniques such as GOSS and EFB to improve the speed without sacrificing accuracy. The model will then be evaluated based on R-squared and RMSE values showing the two model strong predictive abilities. After further tests, it was found that CatBoost is the model that gives the best results. CatBoost is a gradient boosting method chosen for its strong ability to model nonlinear relationships and feature interactions.

Finally, the primary deliverable of this project is an analysis of drug efficacy patterns and the identification of possible relational features, such as genes that may influence drug response. This analysis could support the development of an interactive dashboard where healthcare professionals and researchers input relevant data to generate possible IC50 predictions. The dashboard would allow the user to explore various predicted drug responses for various genetic profiles, view key features that influence treatment results, and compare different effectiveness of various treatment plans. (PLACEHOLDER FOR TRUE FINAL RESULTS)

*Progress*

<u>Data Preparation</u>

This stage focused primarily on converting the Genomics of Drug Sensitivity in Cancer (GDSC) dataset [1] and other associated datasets into a biologically consistent and structurally complete modeling dataset for the LN_IC50 prediction models. Garnett, Mathew J., et al. presented the GDSC platform as a large-scale panel of cancer cell lines screened against anticancer compounds, with genomic characterization that included cancer gene mutation, copy number, gene expression, rearrangements, and microsatellite instability, while drug treatment effects were summarized using parameters that included IC50 [1]. As part of the preprocessing phase, The missing values present in TCGA, Tissue-descriptor, MSI, and other variables were filled using the additional datasets associated with this study[1] by matching the COSMIC_ID of each row, as COSMIC_ID is a universal key. The TCGA_DESC contained the cancer-type labels, but it also included the non-biological category of UNCLASSIFIED, which accounted for nearly 20% of the dataset. This issue was resolved by applying a staged recovery process that cross-filled missing or weak labels from validated supplementary sources, used reviewed cell-line remaps where appropriate, and only replaced UNCLASSIFIED when the supporting evidence pointed to one clear biological category. 3 new numerical features were also added from the supplementary datasets; mutational_burden, ploidy_snp6, and ploidy_wes; and were also imputed to preserve the sample size. The TARGET was removed due to its missing values, which could not be reliably recovered from the supplementary files.

After cleaning and augmentation, the feature-engineering stage transformed the selected predictors into a model-ready format while preserving interpretability *(fig. 1)* [2]. Because IC50-based response had served as a standard endpoint in GDSC analyses and had also been used directly in prior predictive modeling studies built from the same resource, LN_IC50 was retained as the main target variable rather than AUC or Z-score [1]. The data were then divided using an 80/20 train-test split so that later transformations were learned only from the training data and data leakage was avoided. Categorical predictors were one-hot encoded, while the high-cardinality DRUG_NAME feature was target encoded so that drug identity could be represented as a single informative numeric variable without creating an overly sparse feature space. The continuous predictors such as mutational_burden and ploidy_wes were then standardized, which changed feature magnitude while preserving distributional shape *(fig. 2)*. Finally, mutual-information feature selection was used to rank predictors by their relationship with LN_IC50, and the strongest retained variables included the encoded drug feature together with the augmented genomic features, showing that the added biological variables contributed meaningful predictive signal rather than noise *(fig. 3)*.

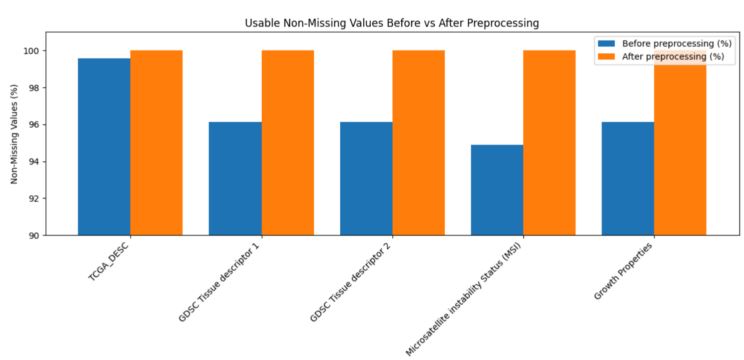


Figure 1 - Percentage of values in key features before and after preprocessing, showing that filling and imputation substantially improved the dataset.

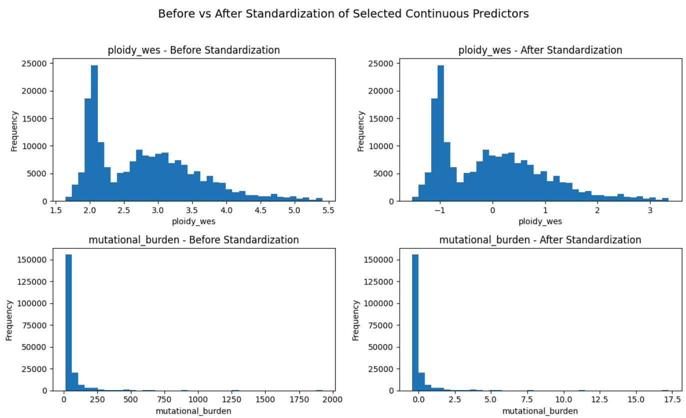


Figure 2 - Distributions of numeric predictors before and after standardization, showing that scaling changed feature magnitude while preserving overall distribution shape.

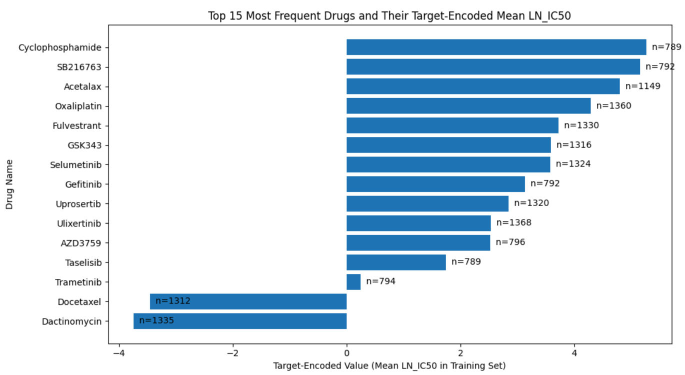


Figure 3 - Top features ranked by mutual information with LN_IC50, highlighting the strongest predictive variables retained in the final feature set.

## Methods

| | Model | RMSE | MAE | R2 |
|---|---|---|---|---|
| 0 | CatBoost | 1.131118 | 0.844668 | 0.832297 |
| 1 | XGBoost | 1.145858 | 0.854324 | 0.827898 |
| 2 | MLPRegressor | 1.189713 | 0.893602 | 0.814473 |
| 3 | HistGradientBoosting | 1.225448 | 0.914313 | 0.803160 |
| 4 | LightGBM | 1.226727 | 0.915228 | 0.802749 |
| 5 | ExtraTrees | 1.249409 | 0.917899 | 0.795387 |
| 6 | Ridge | 1.344436 | 1.008067 | 0.763079 |

Figure 4 - Comparison of model performance with default parameters using RMSE, MAE, and $R^2$ to identify the strongest-performing approach. Models are ranked according to $R^2$.

Six different models were evaluated in their ability to accurately predict LN_IC50 from genetic and molecular data. CatBoost and XGBoost, both gradient boosting methods, were chosen for their strong ability to model nonlinear relationships and feature interactions in molecular descriptor data, with CatBoost additionally offering robustness through ordered boosting. MLPRegressor was included as a neural network approach capable of capturing potentially complex nonlinear mappings, while HistGradientBoosting and LightGBM were chosen as possibilities for their efficient, scalable boosting techniques suited for this large, high-dimensional dataset. ExtraTrees offered a complementary ensemble method that is resistant to overfitting and reduces variance through randomized splits. Ridge regression was used as a baseline due to its simplicity and effectiveness in handling high dimensionality and multicollinearity, although due to its linear nature, did not perform as well as nonlinear models. Across all models, default parameters were used, providing a consistent baseline comparison; however, further hyperparameter tuning is expected to improve predictive accuracy.

Performance

There are three different metrics used to compare models: root mean squared error (RMSE), mean absolute error (MAE), and $R^2$. Both the MAE and RMSE are in the same units as the target variable, making them easily comparable to each other. RMSE penalizes large errors more heavily than the MSE, which is important in the context of this project because large deviations can have a significant impact on the interpretation of biological data and drug potency. The MAE is less sensitive to outliers than the RMSE, so provides a better measure of typical error. $R^2$ is the coefficient of determination; a measure of model fit independent of scale and gives good context for overall model performance. So, the overall interpretation of model performance is mostly determined by $R^2$ and further informed by MAE and RMSE. A well-performing model will have a low RMSE and MAE, with a small difference between the two, and a high $R^2$. By all 3 measures of model performance, CatBoost performed the best, with XGBoost following. Both are tree-based boosting models that handle nonlinear modeling well. There are moderate differences between the MAE and RMSE across all models which indicates the presence of significant outliers in the LN_IC50 data.

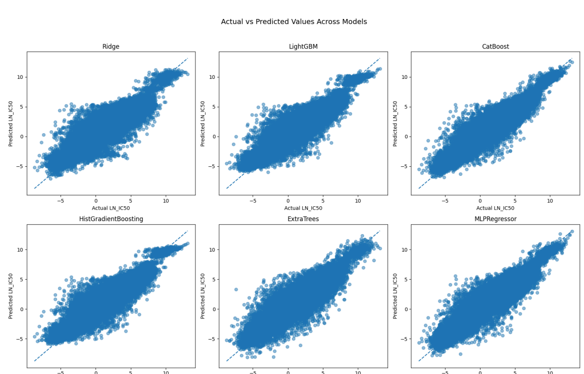


Figure 5: Scatter plots comparing actual vs. predicted values for all 6 models tested.

The actual-versus-predicted plots show that all six models captured the overall positive relationship in LN_IC50 reasonably well, since the points in every panel followed the diagonal trend rather than appearing randomly scattered. CatBoost, MLPRegressor, and ExtraTrees appeared to track the diagonal most tightly, which suggests stronger predictive accuracy, while Ridge showed a wider spread around the line, indicating comparatively larger errors. Across all models, the point clouds were densest in the middle range and became less precise at the extremes, suggesting that very low and very high LN_IC50 values were harder to predict exactly than the central values.

*Plan*

Week 1:

- Try Cross-Validation target encoding to the preprocessing file
- Finalize parameter selection for CatBoost and XGBoost model to improve LN_IC50 predictions

Week 2:

- Write the final interpretation section, explaining what the model has learnt and how the selected features relate to predicting drug responses.
- Finish writing up the documents in GitHub and prepare for the Project Demo.

Week 3:

- Optional: Try to incorporate a Dashboard that could be used by the targeted audience to check the predicted Ln_IC50 score of a drug.

Week 4:

- Finish everything and upload

*Remaining questions*

Remaining questions to wrap up at the end of the project is to finalize the data and graphs, then analyze them. The group will work on the final deliverable and deep analysis with time prevailing, possibly creating a user friendly dashboard to allow patients easy access to their possible adverse side effects from their cancer treatments. It is still in question if there will be time to complete the dashboard. In the article, Menden et al. (2013), there is a major headway for this type of analysis leading to personalized treatments for patients with cancer. Still, research needs to be done regarding the different ways to analyze the data, but as of right now, the group will be focusing on the correlations. The immediate question right now would be choosing the model and its parameters to get the most accurate predictions. Overall, all previous questions have been answered with respect to progress with the analysis still being made.

*Expected results*

The final models are expected to predict LN_IC50 with very strong accuracy and precision, this means values should be reasonably close to the actual drug response values. Based on prior research that was done with the same data sets, such as Menden et al.(2013), and results from similar Kaggle implementations, it is right to expect that the models will capture a substantial portion of the relationship between genomic features and drug sensitivity.

Based on a previous demo, using the default dataset, the LightGBM model is expected to outperform linear models due to its ability to learn more complex and nonlinear patterns that are in the data. Linear models are still expected to provide stable baseline performance, but may not fully capture deeper relationships between variables. This could change once the new preprocessed dataset is run through hyper-tuned parameter models and evaluated using the metrics. For evaluation metrics, the expected performance is: R^2 in the range of 0.65-0.80, which means that a large portion of the variation in the drug response is explained. RMSE ranges from 1.2-1.5, which reflects a moderate prediction error. MAE around 0.9-1.2, indicates the relative low average error.

The expectations above are consistent both in the published research and kaggle submission implementations using similar datasets and models techniques. It is also expected that a subset of features will have a stronger influence on predictions than others, this shows that certain variables contribute more to drug response than others. This allows for clearer identification of the most impactful predictors. Overall, the results are expected to demonstrate reliable predictive performances and reveal meaningful patterns between input features and drug sensitivity.

*References*


1) Garnett, Mathew J., et al. "Systematic identification of genomic markers of drug sensitivity in cancer cells." Nature, vol. 483, no. 7391, Mar. 2012, pp. 570–575, https://doi.org/10.1038/nature11005.
2) Menden, Michael P., et al. "Machine learning prediction of cancer cell sensitivity to drugs based on genomic and Chemical Properties." PLoS ONE, vol. 8, no. 4, 30 Apr. 2013, https://doi.org/10.1371/journal.pone.0061318.
3) Gao, Qing, et al. "The artificial intelligence and Machine Learning in lung cancer immunotherapy." *Journal of Hematology & Oncology*, vol. 16, no. 1, 24 May 2023, https://doi.org/10.1186/s13045-023-01456-y.
4) Cavallari, Larisa H. & Johnson, Julie A. (2023). Use of multi-gene pharmacogenetic panel reduces adverse drug effects. Cell Reports Medicine, 101021. https://www.cell.com/cell-reports-medicine/fulltext/S2666-3791(23)00131-3?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2666379123001313%3Fshowall%3Dtrue
5) (2025). Cancer Cell Line Encyclopedia. Depmap Portal. https://sites.broadinstitute.org/ccle/